

Taxi Booking System



Taxi Booking System

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Project Overview

In cities and localities where taxis are expensive and booking a cab manually takes too much time, the majority of people rely on cabs for transportation. These methods are often slow, inconvenient, and unreliable, especially during busy hours. Passengers frequently experience long waiting times, unclear fare charges, and uncertainty about whether a taxi is even available. At the same time, drivers may have difficulty finding passengers regularly, leading to uneven ride distribution and inefficient use of time and fuel.

The Online Taxi Booking System is a web-based platform that enables clients to book standard and executive taxis conveniently from their homes or workplaces. Using the system, passengers can easily book a taxi, view driver details, track the vehicle's location, and make cashless payments. Drivers can manage their availability, receive ride requests, and complete trips more efficiently. Administrators are able to supervise system operations, manage users, and ensure smooth service delivery. Overall, the system helps improve efficiency, transparency, and user satisfaction in urban taxi services.



Objectives of the project

The main objective of the Online Taxi Booking System is to provide a fast, reliable, and user-friendly platform that simplifies the process of booking taxis and improves the efficiency of taxi services. The system enables passengers to request taxis easily, track assigned drivers, and make cashless payments, while allowing drivers to manage availability and efficiently complete trips. Administrators are provided with tools to monitor system activity, manage users, and maintain service quality. Overall, the system aims to improve efficiency, transparency, and user satisfaction in urban transportation services.

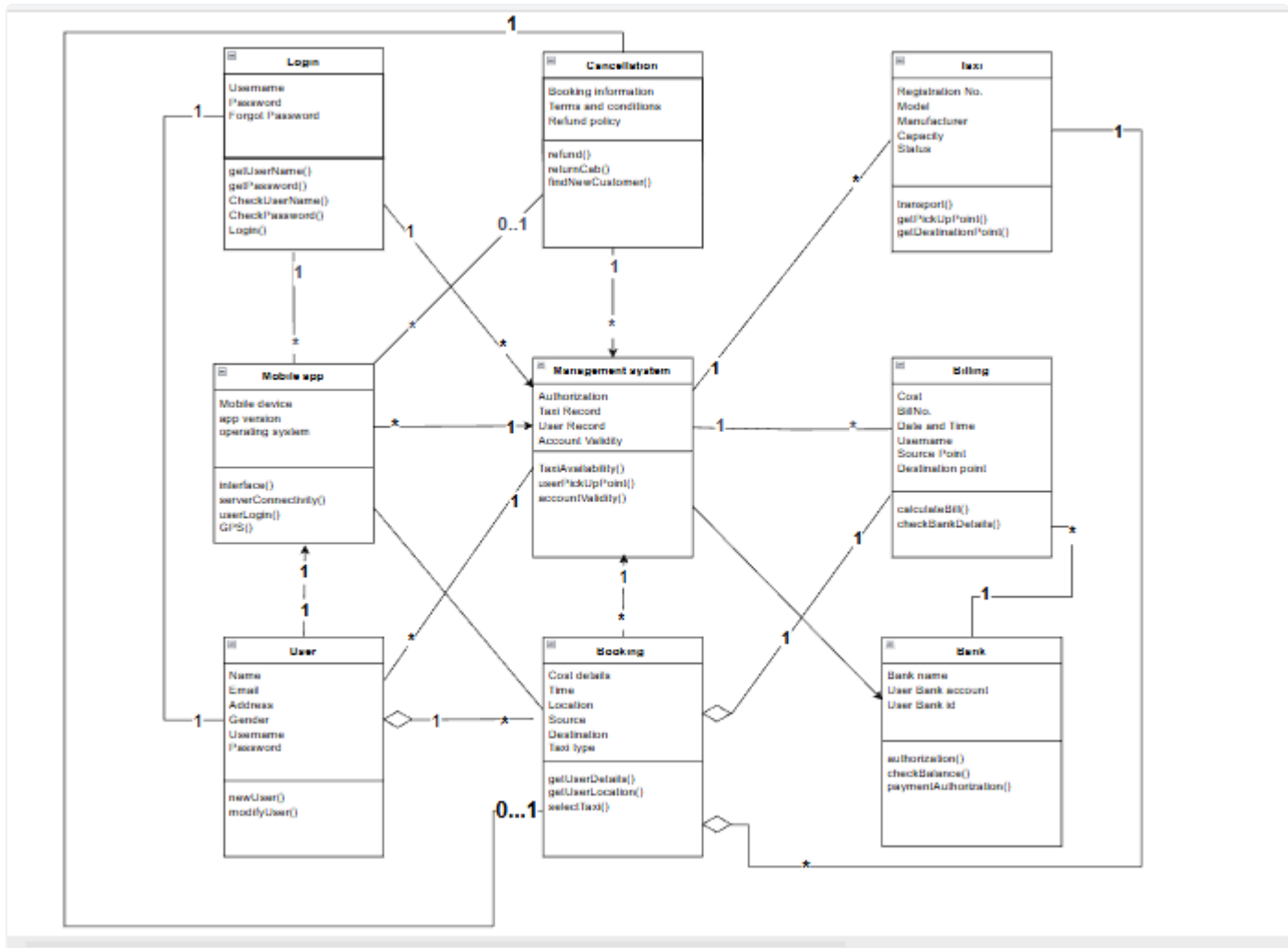
1. Customer / Passenger

Field Name	Description	Data Type	Length
user_id	Unique user identifier	Integer	
name	User's full name	Varchar	50
email	User email address	Varchar	50
username	Login username	Varchar	50
password	Encrypted password	Varchar	50
gender	gender	Varchar	10
address	user address	Varchar	100

2. Booking

Field Name	Description	Data Type	Length
booking_id	Unique booking number	Integer	
user_id	User making booking	Varchar	50
source	Pickup location	Varchar	50
destination	Drop off location	Varchar	50
taxi_type	Type of taxi	Varchar	50
gender	gender	Varchar	10

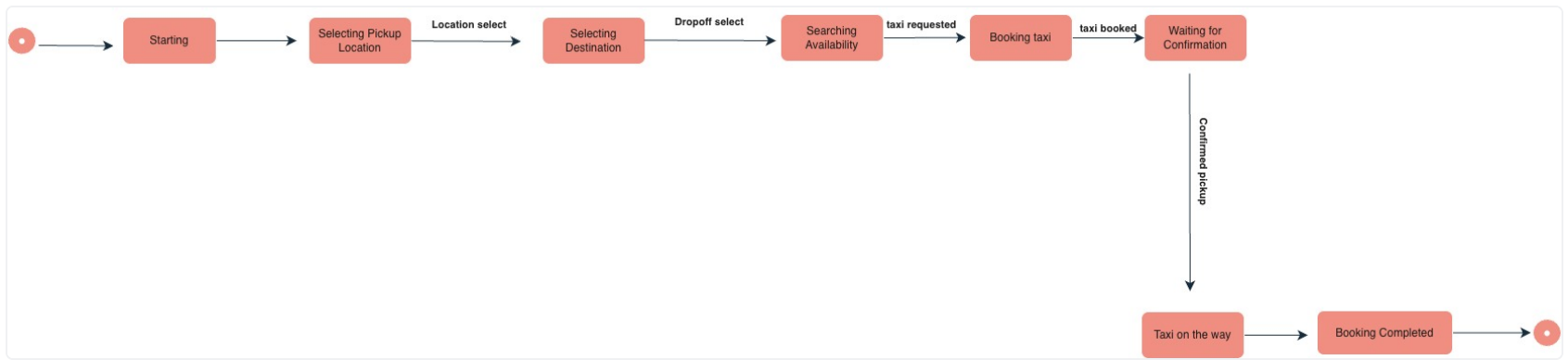
1. Class Diagram



2. State Diagram

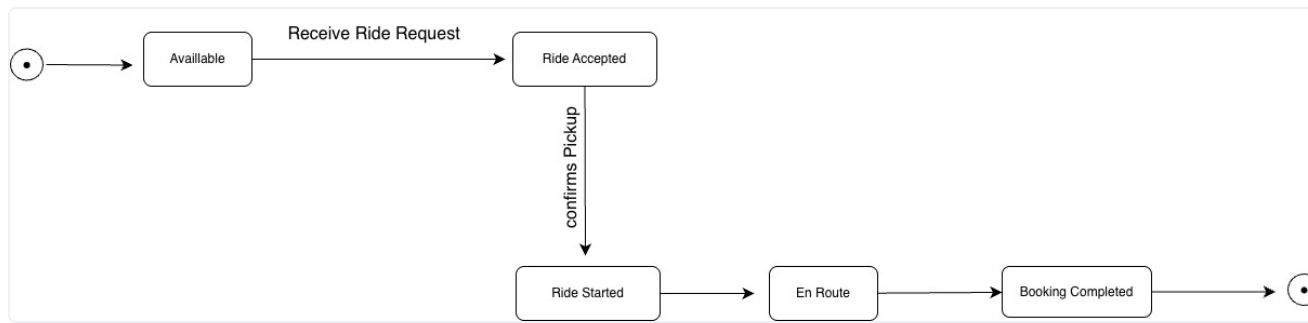
2.1: Customer / Passenger

This state diagram shows how a passenger interacts with the taxi booking system, including the different states the user goes through from booking a taxi to completing a trip.



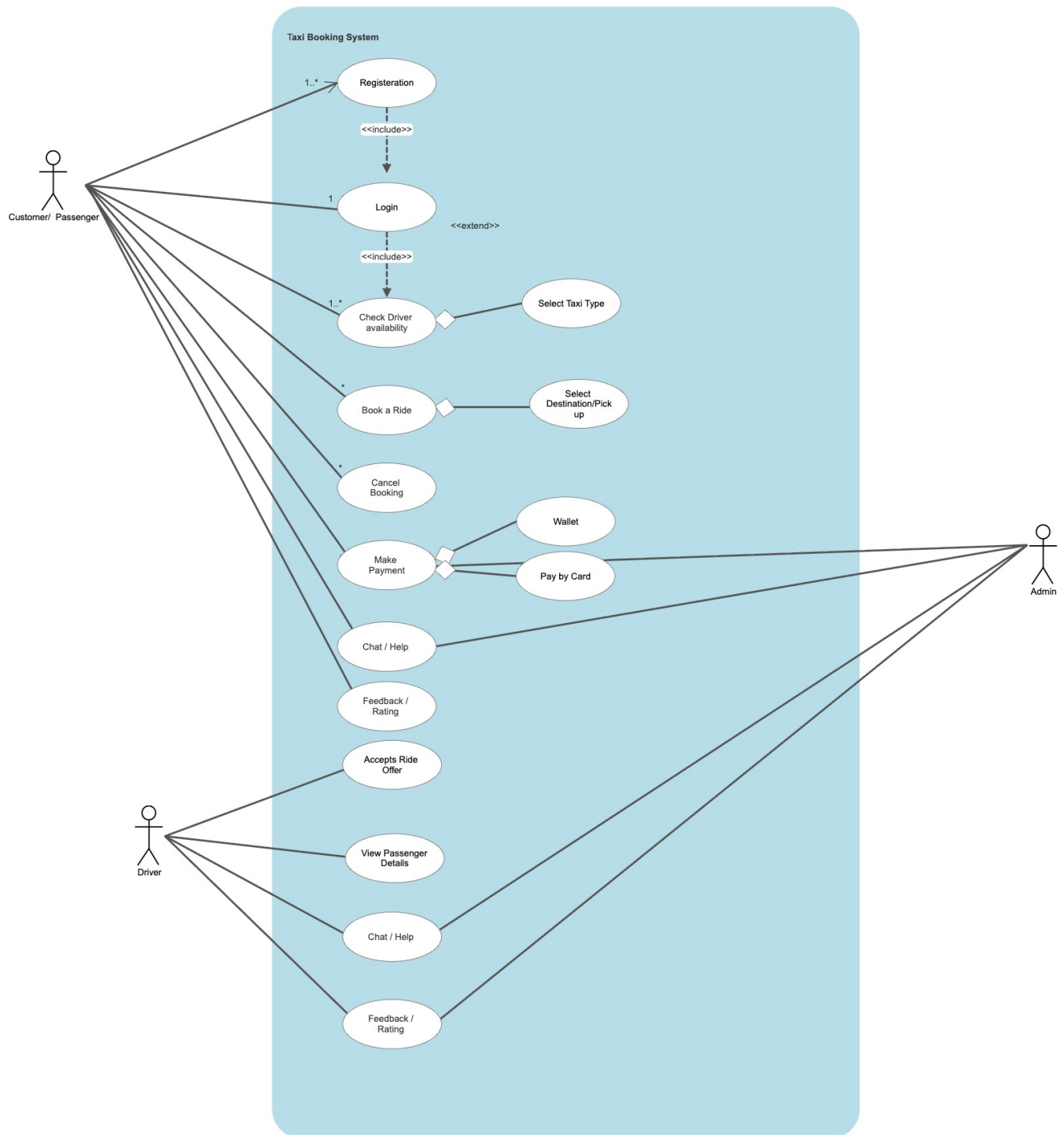
2.2: Driver

This state diagram shows how a driver interacts with the taxi booking system



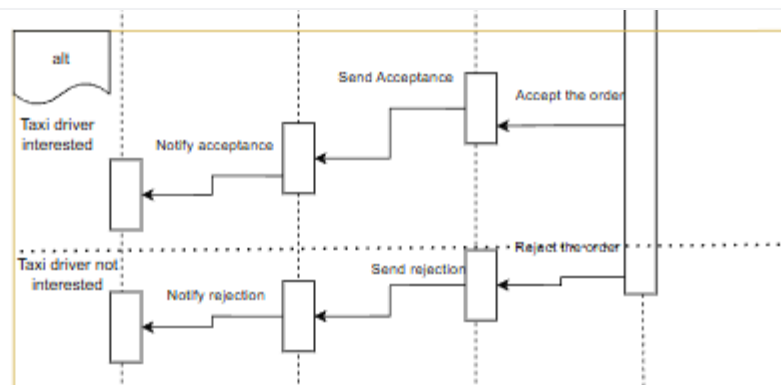
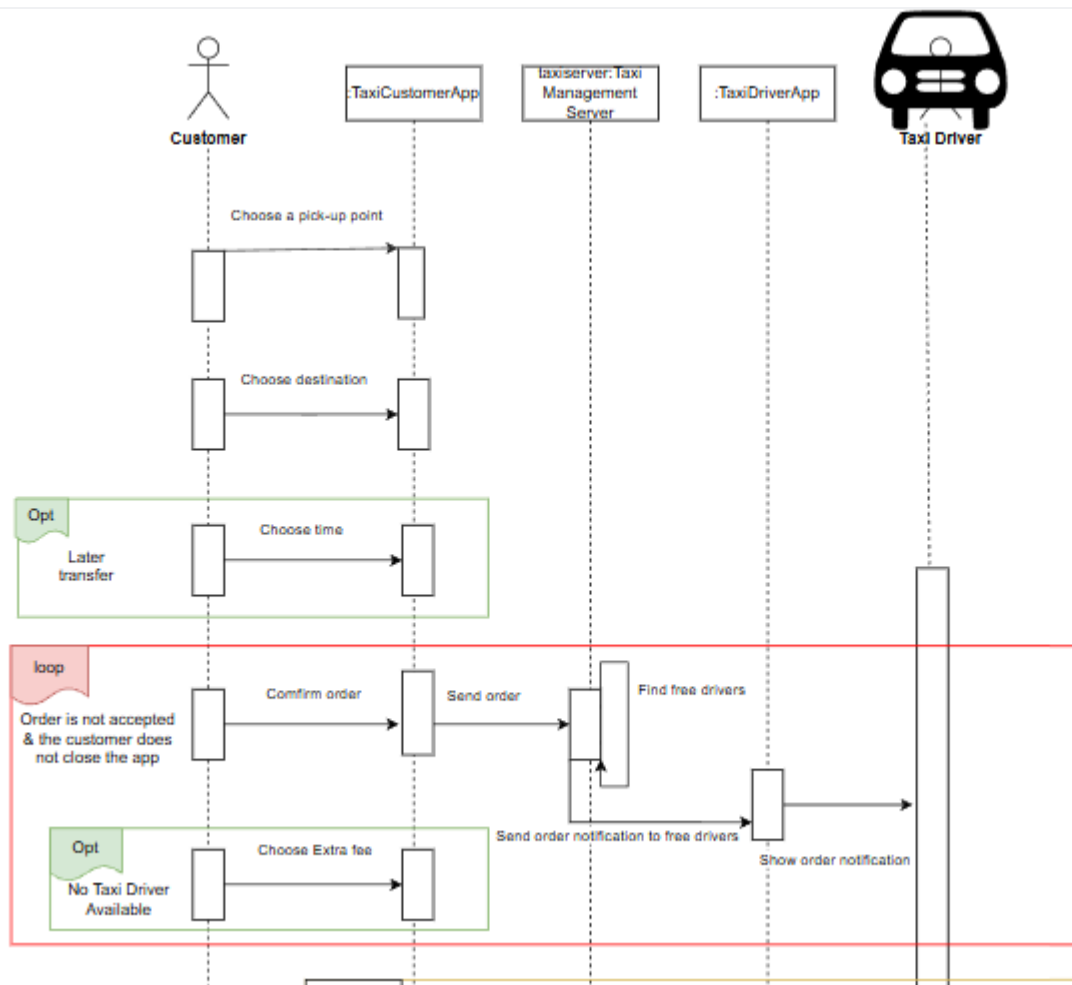
3. Use Case Diagram

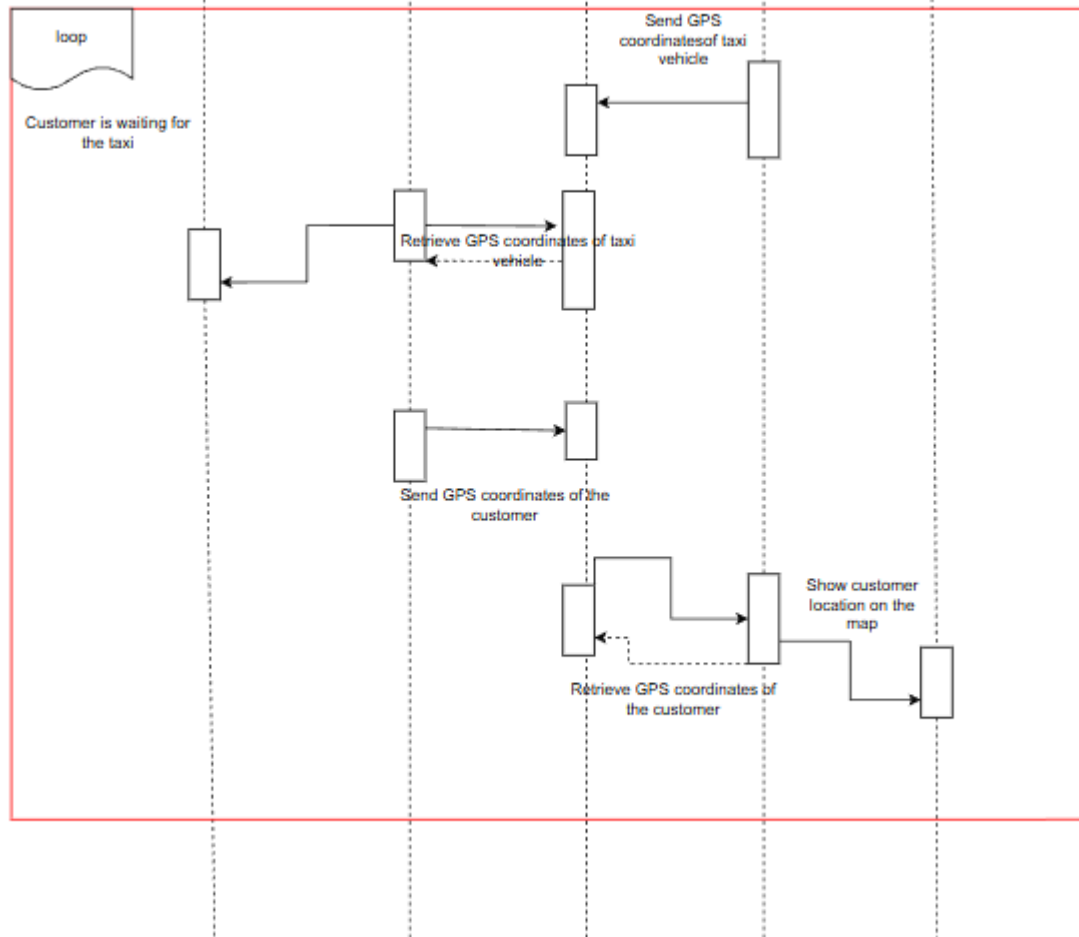
A use case diagram for an online taxi booking system visually represents the interactions between users (actors) and the system. The main actors typically include **Customer**, **Driver**, and **Admin**. Key use cases for the **Customer** include *Register/Login*, *Book Ride*, *Make Payment*, and *Provide Feedback*. For the **Driver**, use cases include *Accept/Reject Ride*, *Update Ride Status*, and *View Passenger details*. The **Admin** manages *User Accounts and Chat/Help*. This diagram helps in understanding the system's functionality and the relationships between different users and their actions.



4. Sequence Diagram

There are an unlimited number of scenarios, so it is not possible to show them all. The basic pattern is established. We tried to elaborate all the use cases and cover the basic kinds of behavior with a sequence diagram.





The diagram shows these lifelines:

- Customer
- Taxi Customer App
- Taxi Management Server
- Taxi Driver App
- Taxi Driver

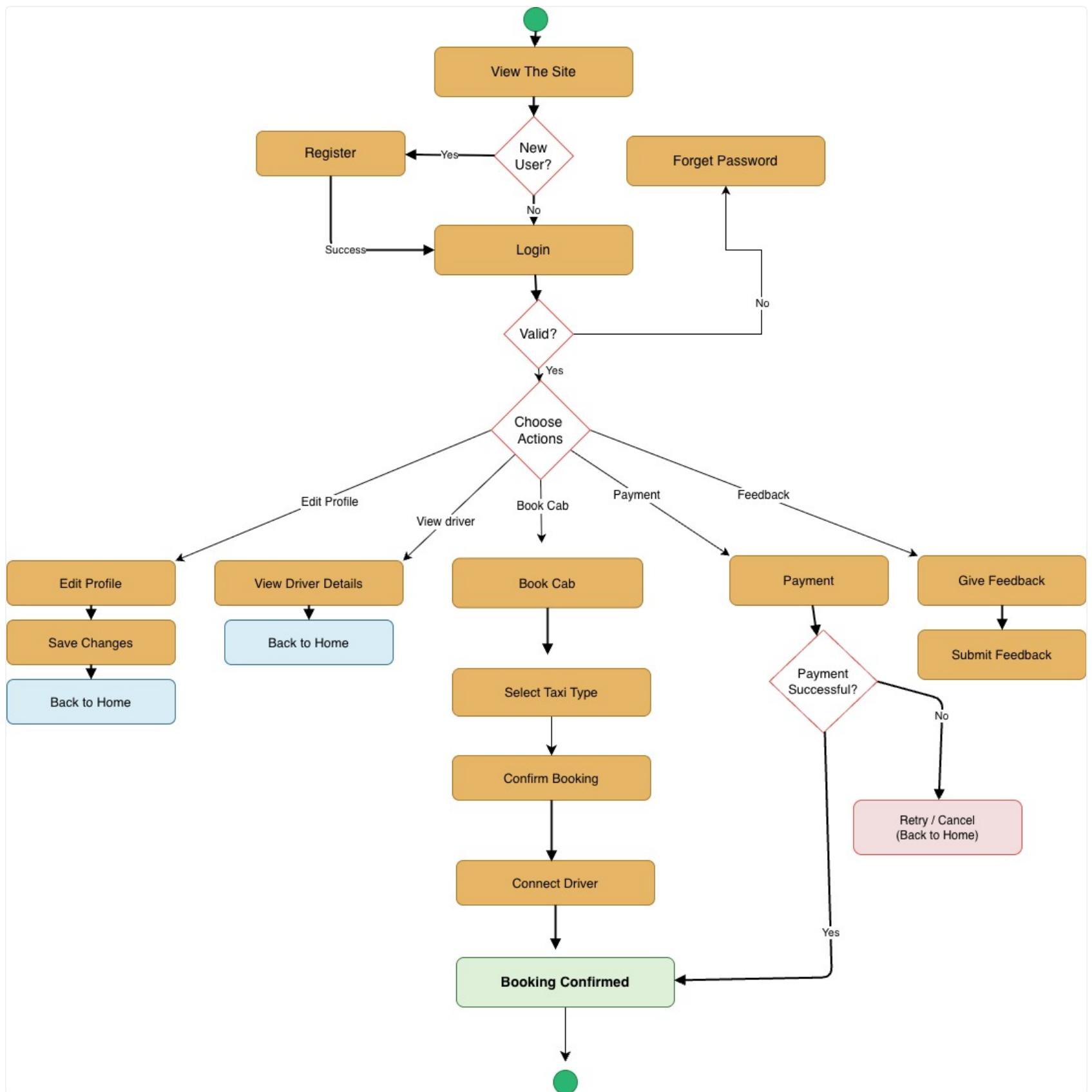
There are these interactions between the lifelines:

- Choose a pick-up point
- Choose destination
- (Optional) Later transfer
 - Choose time
- (Loop) The order is not accepted, and the customer does not close the app
 - Confirm order
 - Send order
 - Find free drivers
 - Send order notification to free drivers
 - Show order notification
 - (Optional) No taxi driver available

- Choose an extra fee
- (Alternative 1) Taxi driver interested
 - Accept the order
 - Send acceptance
 - Notify acceptance
- (Alternative 2) The taxi driver is not interested
 - Reject the order
 - Send rejection
 - Notify rejection
- (Loop) The customer is waiting for the taxi
 - Send the GPS coordinates of the taxi vehicle
 - Retrieve the GPS coordinates of the taxi vehicle
 - Show the location of the taxi vehicle on the map
 - Send the GPS coordinates of the customer
 - Retrieve the GPS coordinates of the customer
 - Show the customer location on the map

5. Activity Diagram

The activity diagram is used correctly to represent the workflow of the system without misusing it as a flowchart substitute. All activities are modeled at a uniform level of abstraction, ensuring clarity and avoiding unnecessary detail. Decision nodes are used properly with clear Yes/No paths, ensuring all conditions lead to valid and complete workflow outcomes. The activity sequence is structured to support understanding, simulation, or potential execution of the process.



Conclusion

Customers can use an online taxi booking system to conveniently rent taxis. Through this platform, users are able to browse available taxis, view driver and vehicle profiles, and book rides according to their needs. Taxi services are a common mode of transportation and are provided by many companies across cities. A large number of people depend on these services for their daily commuting and travel requirements.

